7.8 BACK EMF OF DC MOTOR (E_B)

When the armature of a DC motor rotates in the magnetic field, the armature conductor cuts the magnetic flux. Hence an EMF will be induced in the conductor. According to Faraday's law of electromagnetic induction. This induced EMF acts in opposite direction to the applied voltage and it is referred as the back emf or counter emf E_b.

The back emf is given by,

$$E_b = \frac{\theta ZN}{60} \times \frac{P}{A} \text{ volts}$$

which is same as that emf induced in a DC generator.

The relationship between the current, back emf and the applied voltage for a DC shunt motor is given by $E_b = V - I_a R_a$.

Where,
E_b = back emf in volts
V = supply voltage in volts
I_a =armature current in amperes
R_a =armature resistance in ohms.

The back emf E_b is always less than that of the applied voltage and hence the current is flowing against the direction of back emf.

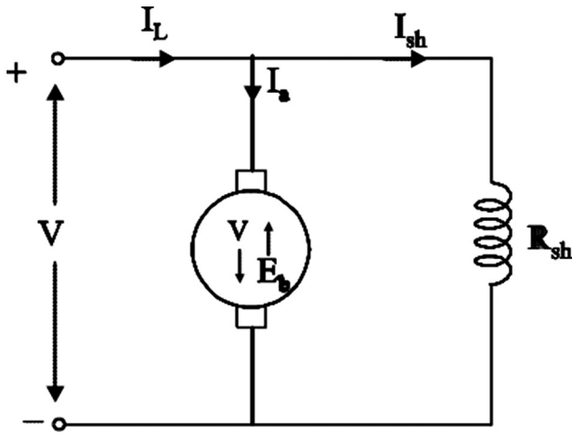


Fig 7.13 *DC Motor Circuit*

7.9 TYPES OF DC MOTORS

The DC motor is classified into three types based on the field winding connection with the armature.

i) DC series motor
ii) DC shunt motor
iii) DC compound motor

DC compound motor is classified as,

a) Long shunt compound motor
b) Short shunt compound motor

7.9.1 DC Series Motor

In DC series motor, the field winding is connected in series with the armature as shown in Fig 7.14. The series field

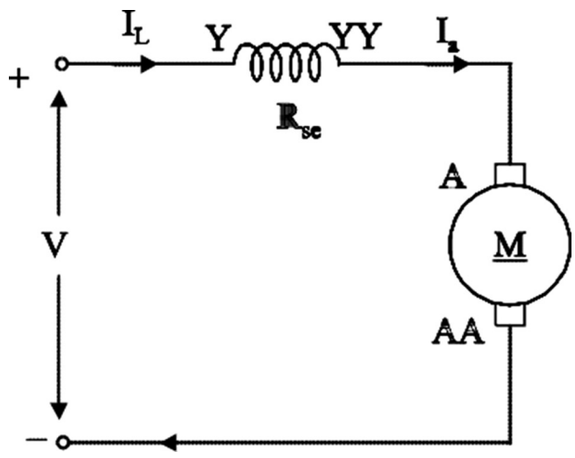


Fig 7.14 *Equivalent Circuit of DC Series Motor*

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